

Code Summary

Store Billing System — C++

Overview

This program simulates a simple department store billing system. It allows a cashier to add items from three different departments — Grocery, Clothing, and Electronics — and calculates the total bill. A 5% loyalty card discount is applied if the customer has one.

Program Structure

Component	Type	Description
grocery_department()	Function	Adds item price to total bill for grocery items
clothing_department()	Function	Adds item price to total bill for clothing items
electronics_department()	Function	Adds item price to total bill for electronics items
main()	Function	Controls program flow, user input, and bill output

How It Works

1. Menu Loop

The program enters a while loop displaying a menu with three department options (Grocery, Clothing, Electronics) and a Finish option (0). The loop continues until the user enters 0.

2. Department Selection

Based on the user's choice, a switch statement routes to the appropriate department function. Each function receives the item price and the total bill variable by reference, adding the price to the running total.

3. Price Accumulation

All three department functions work identically — they add the given price to the total using pass-by-reference (float &total;), so the main total_bill variable is updated directly.

4. Loyalty Card Discount

After finishing, the program asks if the customer has a loyalty card. If yes (y/Y), a 5% discount is deducted from the total and the discounted final bill is displayed. Otherwise, the full total is shown.

Key C++ Concepts Used

Pass-by-Reference: float &total; allows functions to modify the caller's variable directly, avoiding global variables.

while Loop: Keeps the menu running until the user chooses 0 to finish.

switch Statement: Cleanly routes user choices to the correct department function.

char Input: Used to capture the loyalty card response (y/n).

Conditional Discount: An if/else block applies the 5% discount only when the loyalty card is confirmed.

Variables

Variable	Type	Purpose
total_bill	float	Accumulates the total cost of all items
price	float	Stores the price of each item entered by the user
choice	int	Stores the department menu selection
loyalty	char	Stores the loyalty card response ('y' or 'n')